CTEs

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CTE Practice Dataset (PostgreSQL)

EMPLOYEES TABLE

```
CREATE TABLE employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
department_id INT,
manager_id INT,
salary INT,
joining_date DATE
);
```

```
INSERT INTO employees VALUES
(1,'Amit',10,NULL,90000,'2020-01-10'),
(2,'Ravi',10,1,60000,'2021-03-15'),
(3,'Neha',20,NULL,85000,'2019-07-20'),
(4,'Priya',20,3,70000,'2022-02-12'),
(5,'Arjun',30,NULL,95000,'2018-11-05'),
(6,'Meera',30,5,65000,'2021-09-10'),
(7,'Kunal',10,1,55000,'2023-04-18'),
(8,'Divya',20,3,72000,'2024-01-25');
```

DEPARTMENTS TABLE

```
INSERT INTO departments VALUES
(10,'IT'),
(20,'HR'),
(30,'Finance');
```

CUSTOMERS TABLE

```
INSERT INTO customers VALUES
(101,'Rahul','Delhi'),
(102,'Sneha','Mumbai'),
(103,'Karan','Pune'),
(104,'Anita','Delhi');
```

ORDERS TABLE

```
INSERT INTO orders VALUES
(1,101,'2026-01-01',5000),
(2,102,'2026-01-03',7000),
(3,101,'2026-01-05',3000),
(4,103,'2026-01-07',9000),
(5,104,'2026-01-10',4000),
(6,102,'2026-01-12',8000),
(7,103,'2026-01-15',6000);
```

PRODUCTS TABLE

```
INSERT INTO products VALUES
(1,'Laptop','Electronics',60000),
(2,'Phone','Electronics',30000),
(3,'Chair','Furniture',8000),
(4,'Desk','Furniture',15000),
(5,'Tablet','Electronics',25000);
```

CTE QUESTIONS

BEGINNER (1-7)

1. Create a CTE to display employees having salary greater than 70000.
2. Find average salary of all employees using CTE.
3. Show total employees in each department using CTE.
4. Find highest salary employee using CTE.
5. Calculate total sales from orders using CTE.
6. Using CTE and JOIN display employee name, department name, salary.
7. Filter Delhi customers using CTE and display their orders.

INTERMEDIATE (8-14)

8. Find employees earning above department average salary.
9. Find total sales per customer using CTE.
10. Find department with highest average salary.
11. Calculate monthly sales using CTE.
12. Rank employees by salary inside department using CTE + Window Function.
13. Find customers spending more than average spending.
14. Find second highest salary using CTE.

ADVANCED INTERVIEW LEVEL (15-20)

15. Find top-performing department based on total employee salary using multiple CTEs.
16. Find top 3 highest-paid employees from each department using CTE + Window Function.
17. Find customers who placed orders but never purchased Electronics products.
18. Create recursive CTE to generate numbers 1 to 20.
19. Create employee salary report with department average and salary difference.
20. Find employee hierarchy using Recursive CTE with Employee, Manager and Level.